

Problem: 2

a) Solution - properties of strong subadditivity

$$S(ABC) + S(B) \leq S(AB) + S(BC)$$

$$S(ABC) - S(BC) \leq S(AB) - S(B)$$

$$S(BC) - S(ABC) \geq S(B) - S(AB) \text{ - Multiply both side by minus}$$

$$\underbrace{S(A) + S(BC) - S(ABC)}_{= I(A:BC)} \geq \underbrace{S(A) + S(B) - S(AB)}_{= I(A:B)} \text{ - Add } S(A) \text{ both side}$$

$$\underline{\underline{I(A:BC) \geq I(A:B)}}$$

b) Solution

$$I(A:B)_S = S(S_A) + S(S_B) - S(S_{AB})$$

Relationship between relative entropy and mutual information

$$I(A:B)_S = D(S_{AB} \| S_A \otimes S_B)$$

$$\begin{aligned} D(S_{AB} \| S_A \otimes S_B) &= \text{Tr } S_{AB} \log S_{AB} - \text{Tr } S_{AB} \log [S_A \otimes S_B] \\ &= \underbrace{\text{Tr } S_{AB} \log S_{AB}}_{= -S(S_{AB})} - \underbrace{\text{Tr } S_{AB} \log S_A}_{= -S(S_A)} - \underbrace{\text{Tr } S_{AB} \log S_B}_{= -S(S_B)} \end{aligned}$$

$$= -S(S_{AB}) - (-S(S_A)) - (-S(S_B))$$

$$= S(S_A) + S(S_B) - S(S_{AB})$$

Apply monotonicity of relative entropy under CPTP-map

$$D(S_{AB} \| S_A \otimes S_B) \geq D((I_A \otimes E)(S_{AB}) \| (I_A \otimes E)(S_A \otimes S_B))$$

$$\geq D(S'_A C \| S_A \otimes S'_C)$$

$$\underline{\underline{D(S_{AB} \| S_A \otimes S_B) \geq D(S'_A C \| S_A \otimes S'_C)}}$$

$$= I(A:B)_S$$

$$= I(A:C)_{S'}$$

2)

$$I(A:B)_S \geq I(A:C)_S$$

c) Solution

The data processing inequality states that mutual information never increases under local TP - quantum operation

Discarding system C is such a local operation acting on BC. Applying the data processing inequality therefore gives

$$I(A:BC) \geq I(A:B), \text{ which is exactly the monotonicity stated in the first question.}$$

2) Solution

$$|GHZ\rangle_{ABC} = \frac{1}{\sqrt{2}} [|000\rangle_{ABC} + |111\rangle_{ABC}]$$

$$\rho_{ABC} = [|GHZ\rangle_{ABC} \langle GHZ|_{ABC}]$$

$$= \frac{1}{2} [|000\rangle_{ABC} \langle 000|_{ABC} + |111\rangle_{ABC} \langle 111|_{ABC}]$$

$$= \frac{1}{2} [|000\rangle_{ABC} \langle 000|_{ABC} + |000\rangle_{ABC} \langle 111|_{ABC} + |111\rangle_{ABC} \langle 000|_{ABC} + |111\rangle_{ABC} \langle 111|_{ABC}]$$

Q) Solution

$$S(A) = S(BA), \text{ where, } S = - \sum_i p_i \log p_i$$

$$S(B) = - \sum_i p_i \log p_i$$

$$\rho_A = \text{Tr}_{BC} [\rho_{ABC}]$$

$$= \text{Tr}_{BC} [\frac{1}{2} |0\rangle_A \langle 0|_A |00\rangle_{BC} \langle 00|_{BC}] + \text{Tr}_{BC} [\frac{1}{2} |0\rangle_A \langle 0|_A |11\rangle_{BC} \langle 11|_{BC}]$$

$$+ \text{Tr}_{BC} [\frac{1}{2} |1\rangle_A \langle 1|_A |00\rangle_{BC} \langle 00|_{BC}] + \text{Tr}_{BC} [\frac{1}{2} |1\rangle_A \langle 1|_A |11\rangle_{BC} \langle 11|_{BC}]$$

$$= \frac{1}{2} |0\rangle_A \langle 0|_A \text{Tr}_{BC} [|00\rangle_{BC} \langle 00|_{BC}] + \frac{1}{2} |0\rangle_A \langle 0|_A \text{Tr}_{BC} [|11\rangle_{BC} \langle 11|_{BC}]$$

$$+ \frac{1}{2} |1\rangle_A \langle 0|_A \text{Tr}_{BC} [|11\rangle_{BC} \langle 00|_{BC}] + \frac{1}{2} |1\rangle_A \langle 1|_A \text{Tr}_{BC} [|11\rangle_{BC} \langle 11|_{BC}]$$

$$S_A = \frac{1}{2} |0\rangle_A \langle 0|_A \underbrace{|00\rangle_{BC}}_{=1} + \frac{1}{2} |0\rangle_A \langle 1|_A \underbrace{|00\rangle_{BC}}_{=0} + \frac{1}{2} |1\rangle_A \langle 0|_A \underbrace{|11\rangle_{BC}}_{=0} + \frac{1}{2} |1\rangle_A \langle 1|_A \underbrace{|11\rangle_{BC}}_{=1}$$

$$S_A = \frac{1}{2} |0\rangle_A \langle 0|_A + \frac{1}{2} |1\rangle_A \langle 1|_A$$

$$S_A = \frac{1}{2} |0\rangle_A \langle 0|_A + \frac{1}{2} |1\rangle_A \langle 1|_A =$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}$$

$$\begin{aligned} S(A) = S(S_A) &= -[\lambda_1 \log \lambda_1 + \lambda_2 \log \lambda_2] = -[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2}] \\ &= -[\frac{1}{2} [\log 2 - \log 2] + \frac{1}{2} [\log 2 - \log 2]] \\ &= -[\frac{1}{2} (0 - 1) + \frac{1}{2} (0 - 1)] = -[-\frac{1}{2} + (-\frac{1}{2})] \\ &= \frac{2}{2} = 1 \\ &= \underline{\underline{1}} \end{aligned}$$

The same mathematical method

$$S(B) = S(S_B)$$

$$S_B = \text{Tr}_A[S_{ABC}] = \frac{1}{2} |0\rangle_B \langle 0|_B + \frac{1}{2} |1\rangle_B \langle 1|_B$$

$$S_B = \frac{1}{2} |0\rangle_B \langle 0|_B + \frac{1}{2} |1\rangle_B \langle 1|_B$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}$$

$$\begin{aligned} S(B) = S(S_B) &= -[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2}] = -[-\frac{1}{2} - \frac{1}{2}] = -[-\frac{2}{2}] = \frac{2}{2} \\ &= \underline{\underline{1}} \end{aligned}$$

$$S(C) = S(S_C)$$

$$S_C = \text{Tr}_{AB}[S_{ABC}] = \frac{1}{2} |0\rangle_C \langle 0|_C + \frac{1}{2} |1\rangle_C \langle 1|_C$$

$$S_C = \frac{1}{2} |0\rangle_C \langle 0|_C + \frac{1}{2} |1\rangle_C \langle 1|_C //$$

3)

$$\begin{aligned}
 S(C) &= S(S_C) = -\left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2}\right] = -\left[-\frac{1}{2} - \frac{1}{2}\right] = -\left[-\frac{2}{2}\right] \\
 &= \frac{2}{2} \\
 &= \underline{\underline{1}}
 \end{aligned}$$

b) Calculate the mutual information $I(A:B)$

Solution

$$S_{AB} = \text{Tr}_C [\rho_{ABC}]$$

$$\begin{aligned}
 &= \text{Tr}_C \left[\frac{1}{2} |00\rangle_{AB} \langle 00|_C + |00\rangle_{AB} \langle 01|_C \right. \\
 &\quad \left. + |11\rangle_{AB} \langle 11|_C + |11\rangle_{AB} \langle 10|_C \right] + \text{Tr}_C \left[\frac{1}{2} |00\rangle_{AB} \langle 01|_C \right. \\
 &\quad \left. + |11\rangle_{AB} \langle 10|_C \right] + \text{Tr}_C \left[\frac{1}{2} |11\rangle_{AB} \langle 11|_C + |11\rangle_{AB} \langle 10|_C \right] + \\
 &\quad \text{Tr}_C \left[\frac{1}{2} |11\rangle_{AB} \langle 10|_C + |11\rangle_{AB} \langle 11|_C \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} |00\rangle_{AB} \langle 00|_{AB} \text{Tr}_C [|0\rangle_C \langle 0|_C] + \frac{1}{2} |00\rangle_{AB} \langle 11|_{AB} \text{Tr}_C [|0\rangle_C \langle 1|_C] \\
 &\quad + \frac{1}{2} |11\rangle_{AB} \langle 00|_{AB} \text{Tr}_C [|1\rangle_C \langle 0|_C] + \frac{1}{2} |11\rangle_{AB} \langle 11|_{AB} \text{Tr}_C [|1\rangle_C \langle 1|_C] \\
 &= \frac{1}{2} |00\rangle_{AB} \langle 00|_{AB} + \frac{1}{2} |11\rangle_{AB} \langle 11|_{AB} = \underline{\underline{\rho_{AB}}}
 \end{aligned}$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}, \quad \lambda_3 = \lambda_4 = 0$$

$$\begin{aligned}
 S(AB) &= S(\rho_{AB}) = -\left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2}\right] = -\left[-\frac{1}{2} - \frac{1}{2}\right] \\
 &= -\left[-\frac{2}{2}\right] = \frac{2}{2} = \underline{\underline{1}}
 \end{aligned}$$

$$\begin{aligned}
 I(A:B) &= S(A) + S(B) - S(AB) \\
 &= 1 + 1 - 1 = 2 - 1 = 1 \\
 &= \underline{\underline{1}}
 \end{aligned}$$

c) Solution ρ_{AB}

$$S_{AB} = \sum_{i=0}^3 p_i S_A^{(i)} \otimes S_B^{(i)}$$

Apply trace: $S_A = \text{Tr}_B [S_{AB}]$

$$S_{AB} = \frac{1}{2} |00\rangle_{AB} \langle 00|_{AB} + \frac{1}{2} |11\rangle_{AB} \langle 11|_{AB}$$

$$S_A = \text{Tr}_B [S_{AB}]$$

$$= \text{Tr}_B \left[\frac{1}{2} |00\rangle_{AB} \langle 00|_{AB} + \frac{1}{2} |11\rangle_{AB} \langle 11|_{AB} \right]$$

$$= \frac{1}{2} |0\rangle_A \langle 0|_A \underbrace{\text{Tr}_B [|0\rangle_B \langle 0|_B]}_{=1} + \frac{1}{2} |1\rangle_A \langle 1|_A \underbrace{\text{Tr}_B [|1\rangle_B \langle 1|_B]}_{=1}$$

$$= \frac{1}{2} |0\rangle_A \langle 0|_A + \frac{1}{2} |1\rangle_A \langle 1|_A$$

$$S_B = \text{Tr}_A [S_{AB}]$$

$$= \frac{1}{2} |0\rangle_B \langle 0|_B + \frac{1}{2} |1\rangle_B \langle 1|_B$$

Therefore,

$S(S_A)$, $S(S_B)$ and $S(S_{AB})$

$$\text{For, } S_A = \frac{1}{2} |0\rangle_A \langle 0|_A + \frac{1}{2} |1\rangle_A \langle 1|_A$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}$$

$$S(A) = S(S_A) = - \left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right] = - \left[\frac{1}{2} (-1) + \frac{1}{2} (-1) \right] = - \left[-\frac{1}{2} - \frac{1}{2} \right]$$

$$= \frac{2}{2} = 1$$

$$\text{For, } S_B = \frac{1}{2} |0\rangle_B \langle 0|_B + \frac{1}{2} |1\rangle_B \langle 1|_B$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}$$

$$S(B) = S(S_B) = - \left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right] = - \left[\frac{1}{2} (-1) + \frac{1}{2} (-1) \right]$$

$$= - \left[-\frac{1}{2} \right] = \frac{2}{2} = 1$$

$$S(B) = S(S_B) = 1$$

$$\text{For, } S_{AB} = \frac{1}{2} |00\rangle_{AB} \langle 00|_{AB} + \frac{1}{2} |11\rangle_{AB} \langle 11|_{AB}$$

$$\lambda_1 = \lambda_2 = \frac{1}{2}, \quad \lambda_3 = \lambda_4 = 0$$

$$S(AB) = S(SAB) = - \left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} + 0 + 0 \right] = - \left[-\frac{1}{2} - \frac{1}{2} + 0 + 0 \right] \\ = - \left[-\frac{2}{2} \right] = \frac{2}{2} = 1$$

$$I(A:B) = S(SA) + S(SB) - S(SAB) \\ = 1 + 1 - 1 \\ = 2 - 1 \\ = 1 > 0$$

$$I(A:B) = 1 > 0$$

3) a)

solution

$$|\psi\rangle_{ABC} = \sum_{k=0}^4 c_k |k\rangle_A \otimes |k\rangle_B \otimes |k\rangle_C, \quad |k\rangle = |k \bmod 3\rangle$$

First calculate

$$SABC = |\psi\rangle_{ABC} \langle \psi|_{ABC} \\ = \sum_{k=0}^4 \sum_{k'=0}^4 c_k c_{k'}^* (|k\rangle \langle k'|_A) \otimes (|k\rangle \langle k'|_B) \otimes (|k\rangle \langle k'|_C)$$

Then, calculate, S_A, S_B, S_{AB}

$$S_{AB} = \text{Tr}_C(SABC) \\ = \text{Tr}_C \left[\sum_{k=0}^4 \sum_{k'=0}^4 c_k c_{k'}^* (|k\rangle \langle k'|_A) \otimes (|k\rangle \langle k'|_B) \otimes (|k\rangle \langle k'|_C) \right]$$

$$S_{AB} = \sum_{k, k'=0}^4 \delta_{k \bmod 3, k' \bmod 3} |k\rangle \langle k'|_A \otimes |k\rangle \langle k'|_B$$

$$\text{where, } \langle k'|_C |k\rangle_C = \delta_{k \bmod 3, k' \bmod 3}$$

to calculate S_A , trace out B from S_{AB}

$$S_A = \text{Tr}_B(S_{AB}) = \text{Tr}_B \left[\sum_{k, k'=0}^4 \delta_{k \bmod 3, k' \bmod 3} |k\rangle \langle k'|_A \otimes |k\rangle \langle k'|_B \right]$$

6)

$$S_A = \sum_{k, k'=0}^4 \delta_{k \bmod 3, k' \bmod 3} |k\rangle \langle k'| \otimes |B_k\rangle \langle B_{k'}| C_k C_k^\dagger$$

$$= \sum_{k, k'=0}^4 \delta_{k \bmod 3, k' \bmod 3} \frac{1 + e^{\frac{2\pi i}{5}(k-k')}}{2} |k\rangle \langle k'|$$

where, $\langle B_{k'} | B_k \rangle = \frac{1 + e^{\frac{2\pi i}{5}(k-k')}}{2}$

The same for S_B

$$S_B = \text{Tr}_A [S_{AB}]$$

$$= \text{Tr}_A \left[\sum_{k, k'=0}^4 C_k C_k^\dagger \delta_{k \bmod 3, k' \bmod 3} |k\rangle \langle k'| \otimes |B_k\rangle \langle B_{k'}| \right]$$

$$= \sum_{k, k'=0}^4 C_k C_k^\dagger \delta_{k \bmod 3, k' \bmod 3} |B_k\rangle \langle B_{k'}|$$

$$= \sum_{k, k'=0}^4 C_k C_k^\dagger \delta_{k, k'} \delta_{k \bmod 3, k' \bmod 3} |B_k\rangle \langle B_{k'}|$$

a) Compute, $I(A:B)$

$$S_{AB} = \sum_{k, k'=0}^4 C_k C_k^\dagger \underbrace{\delta_{k \bmod 3, k' \bmod 3}}_{= \delta_{k, k'}} |k\rangle \langle k'| \otimes |B_k\rangle \langle B_{k'}|$$

S_{AB} - splits into three orthogonal blocks (by the value of $k \bmod 3$)
these are proportional

$$r_0 = |C_0|^2 + |C_3|^2 = |0.3|^2 + |0.5|^2 = 0.34$$

$$r_1 = |C_1|^2 + |C_4|^2 = |0.4|^2 + |0.5|^2 = 0.41$$

$$r_2 = |C_2|^2 = |0.5|^2 = 0.25$$

$$S(S_{AB}) = H(r)$$

$$= -[0.34 \log_2(0.34) + 0.41 \log_2(0.41) + 0.25 \log_2(0.25)]$$

$$= -[0.34[-1.5564] + 0.41[-1.2863] + 0.25[-2]]$$

7)

$$= 0.5292 + 0.5274 + 0.1$$

$$= 1.1566$$

2) SB and S(SB)

$$S_B = \sum_{k=0}^4 |C_k|^2 |P_k\rangle\langle P_k|$$

$$\text{where, } |P_k\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\theta_k} |1\rangle), \quad \theta_k = \frac{2\pi k}{5}$$

$$|P_k\rangle\langle P_k| = \frac{1}{2} \begin{bmatrix} 1 & e^{-i\theta_k} \\ e^{i\theta_k} & 1 \end{bmatrix}$$

3) have a clue, 1 $P_k = |C_k|^2$

$$P_0 = 0.09$$

$$P_1 = 0.16$$

$$P_2 = 0.25$$

$$P_3 = 0.21$$

$$P_4 = 0.29$$

All probabilities

Σ

$$\underline{\underline{\sum_k P_k = 1}}$$

$$S_B = \sum_{k=0}^4 P_k \cdot \frac{1}{2} \begin{bmatrix} 1 & e^{-i\theta_k} \\ e^{i\theta_k} & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sum_k P_k & \sum_k P_k e^{-i\theta_k} \\ \sum_k P_k e^{i\theta_k} & \sum_k P_k \end{bmatrix}$$

where,

$$\sum_k P_k = 1$$

$$S = \sum_{k=0}^4 P_k e^{i\theta_k}$$

$$S_B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} S^* \\ \frac{1}{2} S & \frac{1}{2} \end{bmatrix}$$

then, apply final steps

8)

$$\lambda = \frac{1}{2} \pm \frac{1}{2} \sqrt{5}$$

$$\sqrt{5} = 0.206349691$$

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{1}{2} \times 0.206349691$$

$$= 0.5 \pm 0.10319846$$

$$\lambda(SB) = \{0.60319846, 0.39680154\}$$

3) S_A and $S(SA)$

Apply same method like S_B and $S(SB)$

$$\lambda(SA) = \{0.28145078, 0.2625842, 0.21, 0.22854962, 0.0774158\}$$

Therefore, for e

$$S(SB) = 2.5566$$

$$\begin{aligned} S(SB) &= -[0.60319846 \log_2 0.60319846 + 0.39680154 \log_2 0.39680154] \\ &= -[-[0.4399098312 + 0.5291310082]] \\ &= 0.9690408394 \\ &\quad \underline{\underline{2.5566}} \end{aligned}$$

$$\begin{aligned} S(SA) &= -\sum_{i=1}^5 \lambda_i \log_2 \lambda_i \\ &= 2.2877865265 \\ &\quad \underline{\underline{2.2877865265}} \end{aligned}$$

9)

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$$

$$= 2.18778653 + 0.6904884 - 2.5565584$$

$$= 3.15683537 - 2.5565584$$

$$= 2.60027692$$

$$\approx \underline{\underline{2.6003}}$$

b) Solution

$$S(\rho_{AB}) = S(\rho_{AB}) = \frac{1}{2} S_{AB} + \frac{1}{2} (I_A \otimes X_B) S_{AB} (I_A \otimes X_B)$$

$$I(A:B) = S(\rho_B) + S(\rho_B) - S(\rho_{AB})$$

1) ρ_A

$$\rho_A = \text{Tr}_B(\rho_{AB})$$

$$= \frac{1}{2} \text{Tr}_B(S_{AB}) + \frac{1}{2} \text{Tr}_B((I \otimes X) S_{AB} (I \otimes X)) = S_A$$

$$= S_A$$

$$S(\rho_A) = S(\rho_A) = 2.2877865359$$

2) ρ_B

$$\rho_B = \text{Tr}_A(\rho_{AB}) = \frac{1}{2} S_B + \frac{1}{2} X S_B X$$

where i

$$\lambda(\rho_B) = \{0.4060424, 0.5939576\}$$

$$S(\rho_B) = -\sum_i \lambda_i \log_2 \lambda_i$$

$$= 0.97440905$$

3) ρ_{AB} and $S(\rho_{AB})$

$$\rho_{AB} = \frac{1}{2} S_{AB} + \frac{1}{2} (I \otimes X) S_{AB} (I \otimes X)$$

$$S(\rho_{AB}) = 2.3642410884 //$$

10)

$$I(A:B) = S(A) + S(B) - S(AB)$$

$$= 2.2877865359 + 0.7744040515 - 2.3642410884$$

$$= 0.7979494911$$

$$\approx \underline{\underline{0.79795}}$$

Problem: 2

1) solution

$$S = \sum_x P_X S_X$$

$$X = S(S) - \sum_x P_X S(S_X)$$

Accessible Entropy Satisfies

$$I(X:Y) \leq X \leq H(X) = H(P_X)$$

$$X = H(X)$$

$$I(X:Y) = H(X)$$

$$I(X:Y) = H(X) - H(X|Y)$$

$$H(X) = H(X) - H(X|Y)$$

$$H(X) - H(X) = -H(X|Y)$$

$$\underline{\underline{H(X|Y) = 0}}$$

Meaning X is a deterministic function of the measurement outcome Y.

Bob can decode X with zero error (perfectly)

$$X = H(P_X)$$

Holevo's bound says

$$I(X:Y) = X$$

1.1)

Pair orthogonal

$$g_x g_{x'} = 0 \text{ for } x \neq x'$$

$$g_x = |x\rangle\langle x|, \text{ for } (\langle x|x'\rangle = 0)$$

S is block-diagonal across the x-subspaces

$$S(g) = H(P_X) + \sum_x P_X S(g_X)$$

$$X = H(P_X)$$

The measurement that perfectly decodes

$$\Pi_x \Pi_{x'} = 0 \text{ (} x \neq x') \text{ , } \sum_x \Pi_x \leq I$$

Povm (measurement)

$$M = \{M_x\}_x \cup \{M_i\}_i$$

$$M_x = \Pi_x$$

$$M_i = I - \sum_x \Pi_x$$

Then every message x

$$\begin{aligned} \Pr(Y=x|X=x) &= \text{Tr}(M_x g_x) \\ &= \text{Tr}(\Pi_x g_x) \\ &= \text{Tr}(g_x) \\ &= \underline{\underline{1}} \end{aligned}$$

for $x' \neq x$

$$\begin{aligned} \Pr(Y=x'|X=x) &= \text{Tr}(M_{x'} g_x) \\ &= \text{Tr}(\Pi_{x'} g_x) \\ &= \underline{\underline{0}} \end{aligned}$$

So the outcome equals the message with certainty

Bob perfectly recovers Alice message

$$\text{If } X = H(P),$$

for perfectly distinguishable (orthogonal supports)

Povm

M_x is not set $\text{supp}(p_x)$ itself; it is the projector onto that support.

2) Solution

Let's say classical variable $X \in \{1, 2, 3, 4\}$

$$\text{Uniform probability } (p_x) = \frac{1}{4}$$

$$g_x = |x\rangle\langle x|$$

Compute the Holevo quantity χ

$$\left. \begin{aligned} g_x &= |x\rangle\langle x| \\ S(g_x) &= 0 \end{aligned} \right\} \text{ for pure state}$$

$$\chi = S(\rho) - \sum_x p_x S(g_x)$$

$$, \quad \rho = \frac{1}{4} \sum_{x=1}^4 g_x$$

$$= S(\rho)$$

Compute S for $|\psi\rangle = |0\rangle$

$$g_x = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

For $|\psi(\phi)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \sqrt{2}e^{i\phi}|1\rangle)$, this covers,

$$\left. \begin{aligned} |\psi_2\rangle \\ |\psi_3\rangle \\ |\psi_4\rangle \end{aligned} \right\} \text{ with } \phi = 0, 2\pi/3, 4\pi/3$$

$$S(\rho) = |\psi(\phi)\rangle\langle\psi(\phi)|$$

$$= \begin{bmatrix} \frac{1}{3} & \frac{\sqrt{2}}{3}e^{-i\phi} \\ \frac{\sqrt{2}}{3}e^{i\phi} & \frac{2}{3} \end{bmatrix}$$

13)

Averaging over

$$\left. \begin{array}{l} \alpha = 0 \\ \alpha = 2\pi/3 \\ \alpha = 4\pi/3 \end{array} \right\} \text{ makes the off-diagonal cancel because}$$

$$1 + e^{i2\pi/3} + e^{i4\pi/3} = 0$$

so,

$$S(\alpha) = \frac{1}{3} (S(0) + S(2\pi/3) + S(4\pi/3))$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1/3 & 0 \\ 0 & 2/3 \end{bmatrix} //$$

Therefore

$$S = \frac{1}{4} S_1 + \frac{3}{4} \begin{bmatrix} 1/3 & 0 \\ 0 & 2/3 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} + \frac{3}{4} \begin{bmatrix} 1/3 & 0 \\ 0 & 2/3 \end{bmatrix} = \begin{bmatrix} 1/4 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 3/12 & 0 \\ 0 & 6/12 \end{bmatrix}$$

$$= \begin{bmatrix} 1/4 + 3/12 & 0 \\ 0 & 6/12 \end{bmatrix} = \begin{bmatrix} 3/12 & 0 \\ 0 & 6/12 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$= \frac{I}{2} \quad , \quad \boxed{\lambda_1 = \lambda_2 = 1/2}$$

$$\chi = S(S)$$

$$= - \left[\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right] = - \left[-\frac{1}{2} - \frac{1}{2} \right] = - \left[-\frac{2}{2} \right] \quad |A|$$

$$X = S(\beta) = \underline{\underline{1}}$$

Ans ended

$$X \leq \log_2 2 = 1$$

or

$$\underline{\underline{X \leq 1 \text{ bit}}}$$

Why the maximum mutual information is strictly less than 1

$$I_{\max}(X:Y) = \max_{\{P_{Y|X}\}} I(X:Y) \leq X$$

To reach 1

we need equality

$$I(X:Y) = X$$

Equality can hold only if all signal states commute

Example:

$$\begin{aligned} \langle \psi_2 | \psi_3 \rangle &= \frac{1}{3} (2 + 2e^{i2\pi/3}) \\ &= \frac{1}{3} (2 - 2 + i\sqrt{3}) \\ &= \frac{i\sqrt{3}}{3} \neq 0 \end{aligned}$$

So, $[\beta_2, \beta_3] \neq 0$ do not commute

$$I_{\max}(X:Y) \leq X = 1 \text{ bit}$$

Therefore the maximum mutual information between Alice and Bob is ≤ 1 bit

b) Source

$$H(X) = \log_2 \frac{4}{3} \approx 0.415 \text{ bit}$$

Constructing the POVM (the "state-elimination" measurement)

signal states, $|\psi_x\rangle$ (with $x = 1, 2, 3, 4$), unique orthogonal null state $|\psi_x^\perp\rangle$

$$\langle \psi_x^\perp | \psi_x \rangle = 0$$

$$\langle \psi^x | \psi^x \rangle = 1$$

$$|\psi_x\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$|\psi_x^\perp\rangle = \begin{pmatrix} -b^* \\ a^* \end{pmatrix}$$

explicit state

$$\begin{aligned} |\psi_1\rangle &= |0\rangle \\ |\psi_2\rangle &= |1\rangle \end{aligned}$$

$$|\psi_2\rangle = \frac{1}{\sqrt{3}} (|0\rangle + \sqrt{2}|1\rangle)$$

$$|\psi_2^\perp\rangle = \frac{1}{\sqrt{3}} (\sqrt{2}|0\rangle - |1\rangle)$$

$$|\psi_3\rangle = \frac{1}{\sqrt{3}} (|0\rangle + \sqrt{2}e^{i2\pi/3}|1\rangle)$$

$$|\psi_3^\perp\rangle = \frac{1}{\sqrt{3}} (\sqrt{2}e^{-i2\pi/3}|0\rangle - |1\rangle)$$

$$|\psi_4\rangle = \frac{1}{\sqrt{3}} (|0\rangle + \sqrt{2}e^{i4\pi/3}|1\rangle)$$

$$|\psi_4^\perp\rangle = \frac{1}{\sqrt{3}} (\sqrt{2}e^{-i4\pi/3}|0\rangle - |1\rangle)$$

4-Outcome POVM

$$M_x = \frac{1}{2} |\psi_x^\perp\rangle \langle \psi_x^\perp|, (x = 1, 2, 3, 4)$$

It is valid POVM

$$\sum_{x=1}^2 M_x = I$$

$$(\sum_x |\psi_x\rangle\langle\psi_x| = 2I)$$

What probabilities POVM give

If Alice sends $|\psi_x\rangle$, Bob gets outcome y with probability

$$\begin{aligned} P(y|x) &= \langle\psi_y|M_y|\psi_x\rangle \\ &= \frac{1}{2} |\langle\psi_y^\perp|\psi_x\rangle|^2 \end{aligned}$$

If $y = x$ then

$$\langle\psi_x^\perp|\psi_x\rangle = 0$$

so,

$$P(x|x) = 0$$

so outcome x eliminates state " x " it can never happen when x was sent.

For $y \neq x$

$$|\langle\psi_y|\psi_x\rangle|^2 = \frac{1}{3}$$

$$\begin{aligned} |\langle\psi_y^\perp|\psi_x\rangle|^2 &= 1 - \frac{1}{3} \\ &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} P(y|x) &= \frac{1}{2} \cdot \frac{2}{3} \\ &= \frac{1}{3} \\ &= \frac{1}{3}, (y \neq x) \end{aligned}$$

so each column of the channel $P(Y|X)$ is

$$(0, 1/3, 1/3, 1/3) \rightarrow \text{up to permutation}$$

So uniform priors $P_X = 1/4$

So uniform output $P_Y = 1/4$

Mutual information achieved

Y - uniform

$$H(Y) = 2 \text{ bits}$$

$$H(Y|X) = \log_2 3 \rightarrow \text{So given}$$

X, Y takes three equally likely values

Thus

$$I(X; Y) = H(Y) - H(Y|X)$$

$$= 2 - \log_2 3$$

$$= \log_2 \frac{4}{3}$$

$$\approx \underline{\underline{0.415 \text{ bits}}}$$

$$I(X; Y) \approx 0.415 \text{ bits}$$

$$\underline{\underline{\quad \quad \quad}}$$

3) Solution

Alice's input alphabet is $\{0, 1, 2\}$
with $P_0 = P_1 = P_2 = 1/3$

$$H(X) = 1 \text{ bit/source symbol}$$

$$R_1 = \log_2 4$$

$$R_2 = \sqrt{1} \log_2 4 + \sqrt{1-1} \log_2 4, 0 \leq \gamma \leq 1$$

15)

Output states

For input $|0\rangle$

$$S'_0 = N(|0\rangle L_0)$$

$$= E_1 |0\rangle L_0 E_1^\dagger + E_2 |0\rangle L_0 E_2^\dagger$$

$$= |0\rangle L_0$$

S'_0 is pure state

$$S(S'_0) = 0$$

For input $|1\rangle$:

$$E_2 |1\rangle = \sqrt{r} |0\rangle + \sqrt{1-r} |1\rangle$$

$$\therefore |\phi\rangle$$

And, $E_1 |1\rangle = 0$

$$E_2 |1\rangle = |0\rangle L_0 |1\rangle$$

$$= |0\rangle (0)$$

$$= 0 \quad \text{hence}$$

$$S'_1 = N(|1\rangle L_1)$$

$$= \underline{\underline{|\phi\rangle L_1}}$$

S'_1 is pure state

$$S(S'_1) = 0$$

Holevo bound

ensemble $\{p_x, S'_x\}$

$$\chi = S(\bar{S}) - \sum_x p_x S(S'_x) \quad , \quad \bar{S} = \sum_x p_x S'_x$$

Here,

$$S(S'_0) = S(S'_1) = 0, \text{ so}$$

$$X = S(\bar{S})$$

$$\bar{S} = \frac{1}{2}(S'_0 + S'_1)$$

In the basis $\{ |0\rangle, |1\rangle \}$

$$S'_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$S'_2 = \begin{pmatrix} r & \sqrt{r(1-r)} \\ \sqrt{r(1-r)} & (1-r) \end{pmatrix}$$

$$\text{so, } \bar{S} = \frac{1}{2} \begin{bmatrix} 1+r & \sqrt{r(1-r)} \\ \sqrt{r(1-r)} & 1-r \end{bmatrix}$$

Eigenvalues are,

$$\lambda_{\pm} = \frac{1 \pm \sqrt{r}}{2}$$

$$\text{Tr}(\bar{S}) = 1$$

$$\det(\bar{S}) = \frac{1-r}{4}$$

$$\sqrt{r^2 - 4\det} = \sqrt{r}$$

Therefore,

$$X = S(\bar{S})$$

$$= H_2\left(\frac{1+\sqrt{r}}{2}\right)$$

where, $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy

Information lost per source symbol

The encoder carries 1 bit/symbol , while Bob can extract at most $X \text{ bits/symbol}$, so the loss is

$$\text{loss} = 1 - X$$

$$= 1 - \frac{1}{2} \left(\frac{1 + \sqrt{r}}{2} \right) \text{ bits per source symbol}$$

